CS & IT ENGINEERING



Algorithms

Sorting Algorithms
Discussion Notes

DPP - 01



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[MCQ]

Imp

#Q. What is the recurrence relation of insertion sort when the array is almost sorted with p element?

A $T(P) = T\left(\frac{P}{2}\right) + O(1)$ ✗ → $\log_2 P$

B $T(P) = T\left(\frac{P}{2}\right) + P^2$ ✗

C $T(P) = 2T\left(\frac{P}{2}\right) + P$ ✗

D $T(P) = 2T\left(\frac{P}{2}\right) + O(1)$ ✓

Best Case

Insertion Sort :

Best Case → Time Order → $O(P)$

Worst Case → $O(P^2)$

Ans :- D

Option: A: $T(P) = T(P/2) + O(1)$

$T(P) = T(P/2) + 1$

$T(P/2) = T(P/2^2) + 1$

$T(P) = T(P/2^2) + 2$

$= T(P/2^3) + 3$

$\vdots$

$T(P) = T(P/2^k) + k$

$= T(1) + \log_2 P$

$= O(\log_2 P)$

$\frac{P}{2^k} = 1$

$2^k = P$

$k = \log_2 P$

B) $T(p) = T(p/2) + p^2$ $\quad$ $O(p)$

$=$

$= a=1,\ b=2,\quad f(p) = p^2$

$\log_b^a = 0$

$\Rightarrow \Omega(p^2)$

C) $T(p) = 2T(p/2) + p \longrightarrow$ Best Case

$T(p/2) = 2T(p/2^2) + p/2$

$T(p) = 2\left[2T(p/2^2) + p/2\right] + p$

$= 2^2 T(p/2^2) + 2 \times \frac{p}{2} + p$

$$T(P) = 2^2 T(P/2^2) + P + P$$

$$= 2^2 T(P/2^2) + 2P$$

$$= 2^3 T(P/2^3) + 3P$$

General Term :

$$T(P) = 2^K T(P/2^K) + K * P$$

For B.C, $P/2^K = 1$

$$2^k = p \implies \boxed{k = \log_2 p}$$

$$T(p) = p * T(1) + p * \log_2 p$$

$$= p * c + p * \log_2 p$$

$$\boxed{T(p) = O(p \log_2 p)}$$

D) $T(P) = 2T(P/2) + 1$ → $O(1)$

$T(P/2) = 2T(P/2^2) + 1$

$T(P) = 2\left[2T(P/2^2) + 1\right] + 1$

$= 2^2 T(P/2^2) + 2 + 1$

$= 2^3 T(P/2^3) + 2^2 + 2^1 + 2^0 \ldots$

General term

$$T(p) = 2^k T(p/2^k) + (2^0 + 2^1 \cdots 2^{k-1})$$

GP $\Rightarrow$ $a = 1$, $r = 2$, $n = k$ $\Rightarrow$

$$\text{Sum} = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{1 \times (2^k - 1)}{2 - 1}$$

$$= (2^k - 1)$$

General Term

$$T(P) = 2^K T(P/2^K) + (2^K - 1)$$

For B.C, $P/2^K = 1$ $\quad 2^K = P$

$$K = \log_2(P)$$

$$T(P) = P * T(1) + (P - 1)$$

$$= P * C + P - 1$$

$$= \boxed{O(P)}$$

[NAT]

#Q. Consider the following A with 8 elements:



60

What is the index value of elements after 3rd pass of selection sort?

Ans:- 6

I/p: 70 60 20 50 40 5 19 21

pass1 :- o/p: 5 | 60 20 50 40 70 19 21

pass2 o/p: 5 19 | 20 50 40 70 60 21

pass 3 o/p: 5 19 20 | 50 40 70 60 21

pass 4 o/p: 5 19 20 21 | 40 70 60 50

Pass 5 o/p: 5 19 20 21 40 | 70 60 50

pass 6 o/p: 5 19 20 21 40 50 | 60 70

pass 7 o/p: 5 19 20 21 40 50 60 70

↳ (n-1) passes always.

[MCQ]

#Q. Consider the following array with 8 elements

50	60	90	65	55	45	85	12

What is result after 3rd pass of bubble sort?

A 50, 60, 55,45, 12, 65, 85, 90

B 12, 45, 50, 60, 90, 65, 55 85

C 90, 85, 65, 50, 60, 55, 45, 12

D 50, 55, 45, 60, 12, 65, 85, 90

Ans: D

3rd pass → max last 3 elem sorted

given: 50 60 90 65 55 45 85 12

↓

pass 1 :- o/p: 50 60 65 55 45 85 12 (90)

pass 2 o/p: 50 60 55 45 65 12 (85) (90)

⇒ pass 3 o/p: 50 55 45 60 12 (65) (85) (90)

↓

pass 4 o/p: 50 45 55 12 (60) (65) (85) (90)

pass 5 o/p: 45 50 12 (55) (60) (65) (85) (90)

pass 6 o/p: 45 12 50 55 60 65 85 90

pass 7 o/p: 12 45 50 55 60 65 85 90

Sorted

(n-1) passes

[NAT] → Numerical

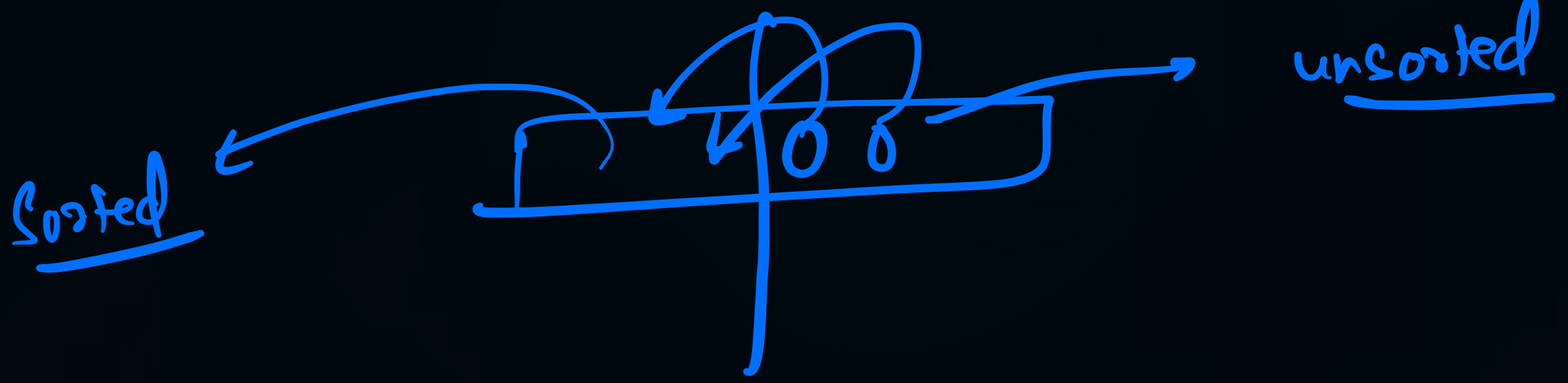
Ans: 16

#Q. Consider the following array A with 9 elements

A

10	20	50	60	70	65	55	25	15

How many swaps are needed to sort the array by using insertion sort___?



Given Input :- 10 20 50 60 70 65 55 25 15

pass 1 : 10 | 20 50 60 70 65 55 25 15
→ 0 swaps

pass 2 : 10 20 | 50 60 70 65 55 25 15
→ 0 swaps

pass 3 : 10 20 50 | 60 70 65 55 25 15
→ 0 swaps

pass 4 : 10 20 50 60 | 70 65 55 25 15
→ 0 swaps

pass 5 : 10 20 50 60 70 | 65 55 25 15
→ 1 swaps

pass 6: 10 20 50 60 65 70 | 55 25 15 → 3 swaps

pass 7: 10 20 50 55 60 65 70 | 25 15 → 5 swaps

pass 8: 10 20 25 50 55 60 65 70 | 15 → 7 swaps

O/p: 10 15 20 25 50 55 60 65 70

Sorted.

Total Swaps: 0 + 1 + 3 + 5 + 7

= 16 swaps

[MSQ]

#Q. Bubble sort is-

Ans :- A, D

A In place sorting technique ✓ → O(1) space

B Outplace sorting technique ✗ → not in place

C Unstable sorting technique ✗

D Stable sorting technique ✓

Stable & Inplace.

[MCQ]

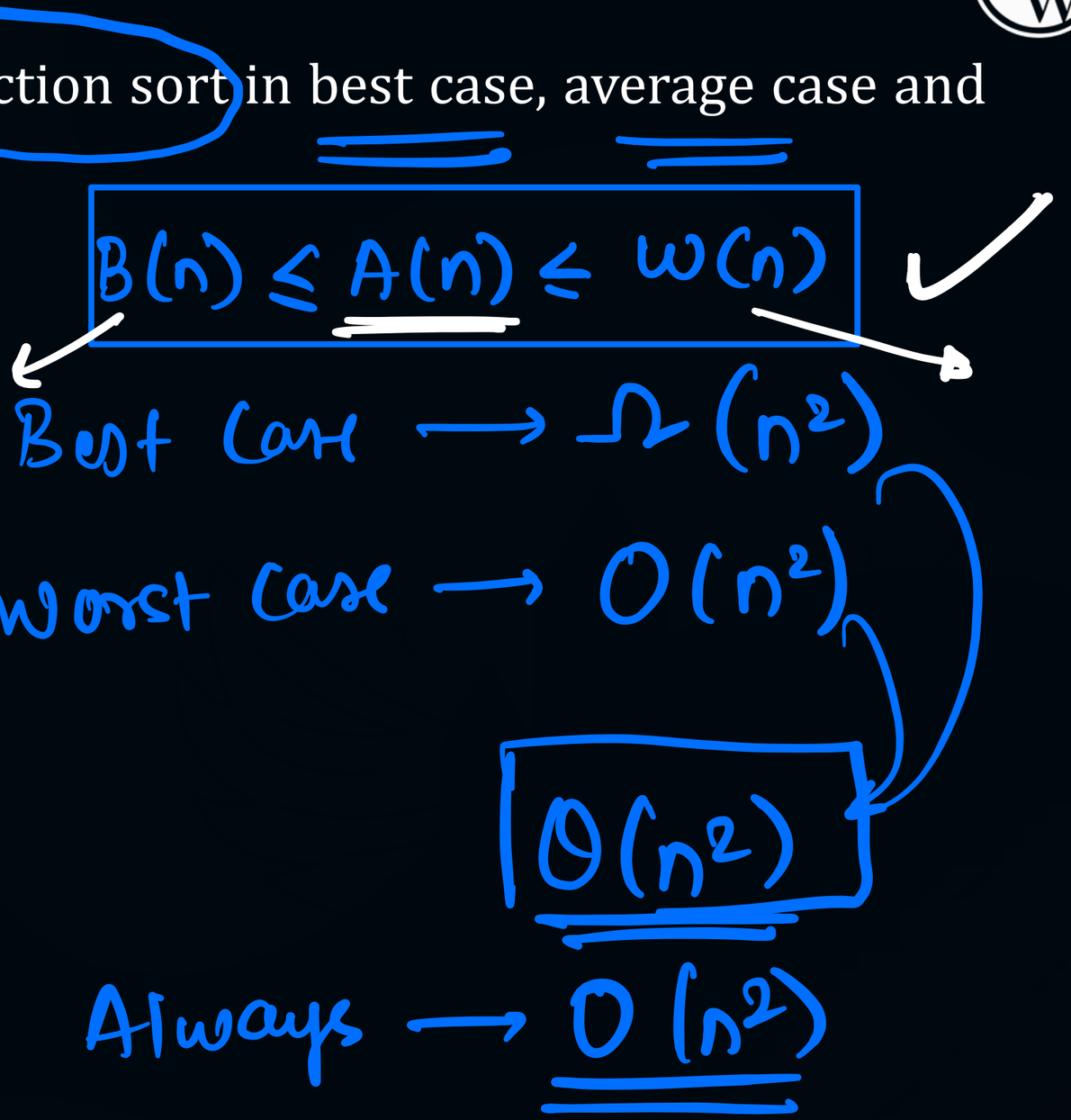
#Q. What is the time complexity of selection sort in best case, average case and worst case respectively is:

A. $O(n), O(n\log n), O(n^2)$

B. $O(n), O(n^2), O(n^2)$

C. $O(n^2), O(n^2), O(n^2)$

D. $O(n), O(n\log n), O(n\log)$



[MCQ]

#Q. How many swaps and needed in selection sort to sort n element in worst case?

↳ takes min swap among all comp algos in use.

Ans:- A

A. n − 1

B. $\frac{n(n-1)}{2}$ ✗

C. n ✗

D. n^2 ✗

Selection Sort ⇒ $\Theta(n^2)$ → Comp + Swaps

Always (n-1)

[MCQ]

#Q. Consider the following elements:

101	56	934	555	8	12	785	23	5	999

What is the result after 3rd pas of radix sort?

A. 56, 23, 12, 8, 5, 101, 555, 785, 934, 999

B. 56, 23, 12, 8, 5, 101, 55, 785, 999, 934

C. 5, 8, 12, 23, 56, 101, 555, 785, 934, 999

D. 5, 8, 12, 23, 56, 101, 555, 785, 934, 999

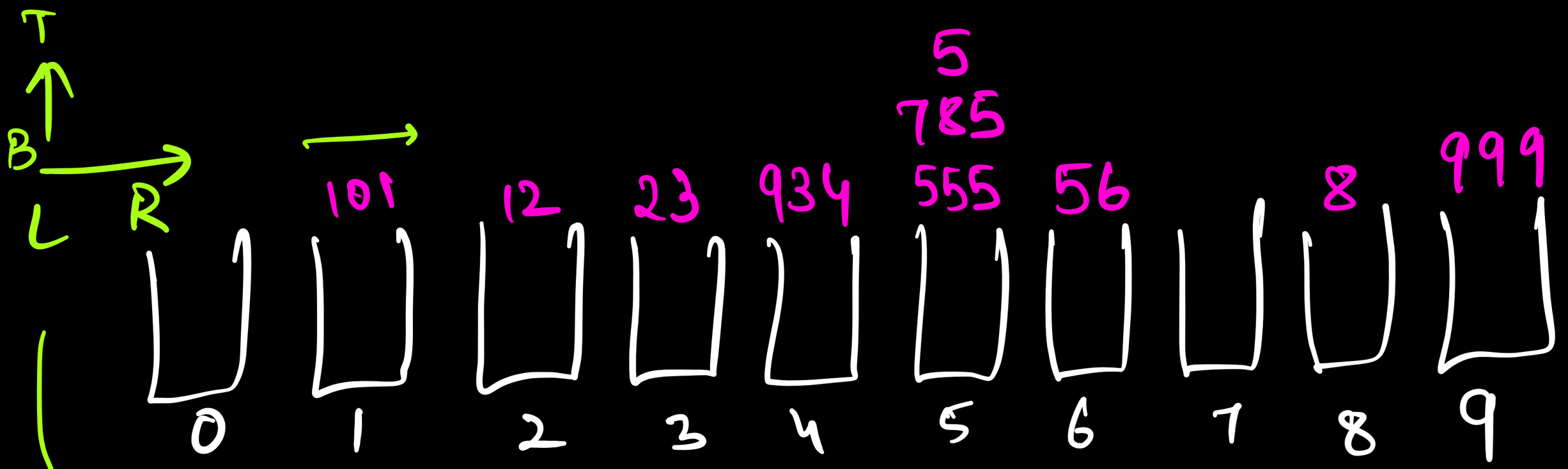
radix Sort
↳ non-comparison
↓
buckets

Ans: D

A: 101 56 934 555 8 12 785 23 5 999

1 6 4 5 8 2 5 3 5 9

Pass1 :- Bucket as per last digit (1st digit from Right)



O/p: Pass1: 101, 12, 23, 934, 555, 785, 5, 56, 8, 999

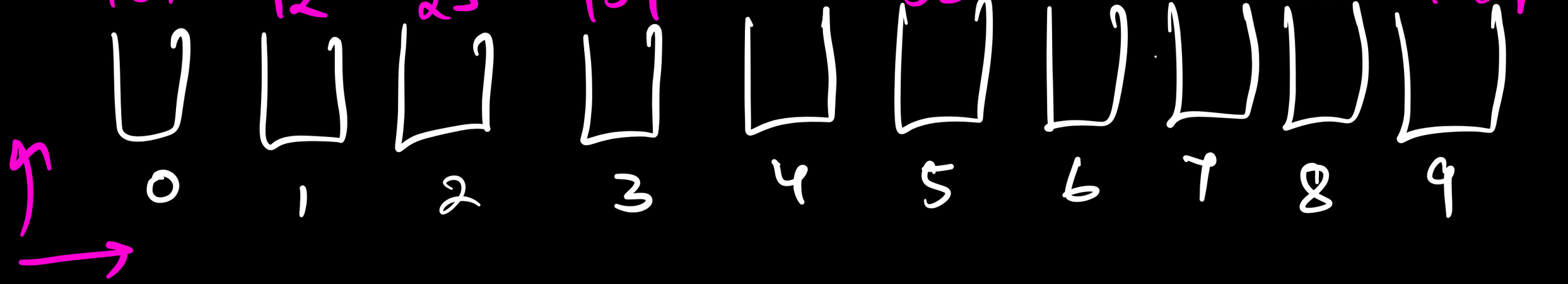
l_0 l_1 l_2 l_3 l_5 l_8 l_0 l_5 l_9

pass 2 : Bucket as per 2nd last digit (2nd digit from right side.

5 → 05 → (0)

8 → 08

8
5
101 12 23 934 56 555 785 999



pass 2 o/p: 101, 5, 8, 12, 23, 934, 555, 56, 785, 999

$\downarrow_0$ $\downarrow_5$ $\downarrow_0$ $\downarrow_7$ $\downarrow_9$

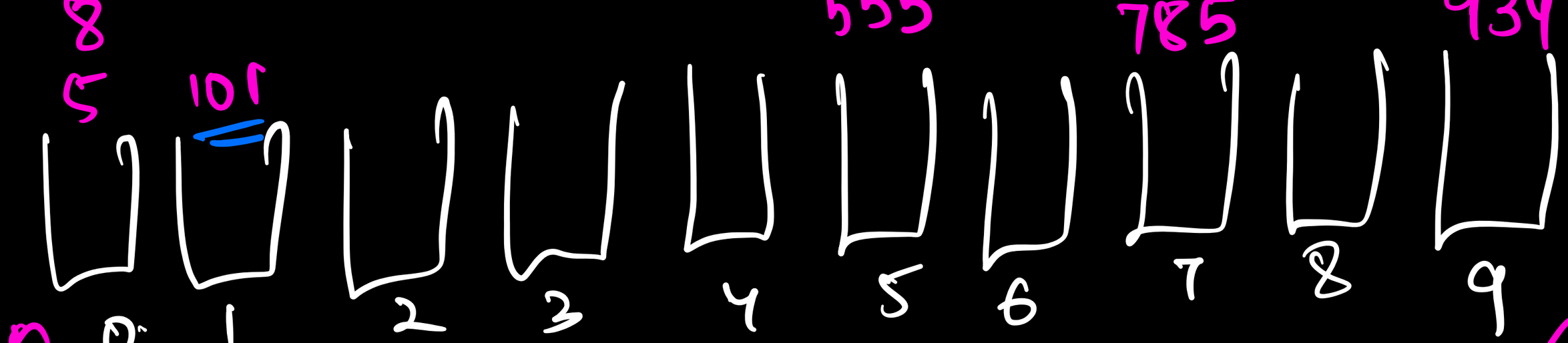
1 0 0

Pass 3: Bucket as per 3rd digit from right side.
(100's place digit)

56
23
12
8

5 → 005
12 → 012

555
785
999
934



→ Pass 3 O/p: → Sorted

5, 8, 12, 23, 56, 101, 555, 785, 934, 999

999



THANK - YOU

→ Lectures
→ DPP/WT
→ Practice
PYQs